Lectures

Stat 154/254: Statistical Machine Learning

Lectures marked with $\star$ are still in progress.

Introduction

- Class introduction
- List of key concepts

Unit 1: Regression in a machine learning setting

- Linear regression in matrix notation (review)
- Linear regression in an ML setting
- Basis expansions and splines for linear regression
- L1 and L2 penalization

Unit 2: Classification

- Empirical risk minimization for classification
- Trading off false positives and negatives
- Perceptron algorithm **(bonus content)**

Unit 3: Risk and complexity

- Asymptotics and bias–variance tradeoff in linear regression
- Regret decomposition
- Cross validation
  - Held-out data and cross-validation
  - Simulated bias–variance tradeoff example
  - Cross–validation cautionary note
- Generalization bounds
  - Generalization error and uniform laws
  - Uniform laws for smooth function classes
  - VC dimension and zero–one loss

Unit 4: Trees and weak learners

- Classification and regression trees
- Bagging weak learners
- Boosting weak learners

Unit 5: Neural networks and optimization

- Optimization notes

Prerequisites

Mathematical statistics, probability, linear algebra, multivariate calculus, and Python are all prerequisites.

For assistance with mathematics prerequisites, the Student Learning Center provides additional assistance.

A summary of the required linear algebra skills can be found here.